

Roadmap towards the redefinition of the SI second

October 6, 2021

Content

1. Introduction - Context.....	2
2. Possible options and scenarios for the redefinition	3
3. Priority setting of criteria and conditions to change the definition	5
4. Mandatory criteria	7
5. Ancillary conditions.....	8
6. Fulfilment levels of criteria and conditions	9
7. Conclusion	12
ANNEX A - Details on mandatory criteria and ancillary conditions - Fulfilment and progress indexes ...	13
ANNEX B - References.....	20

1. Introduction - Context

Since 1967, the definition of the SI second has relied on the caesium atom hyperfine transition frequency. Caesium primary frequency standards currently realize this unit with a relative frequency uncertainty at the 10^{-16} level, but in recent years they have been surpassed by optical frequency standards (OFS) with much lower uncertainties, typically 2 orders of magnitude better (Figure 1).

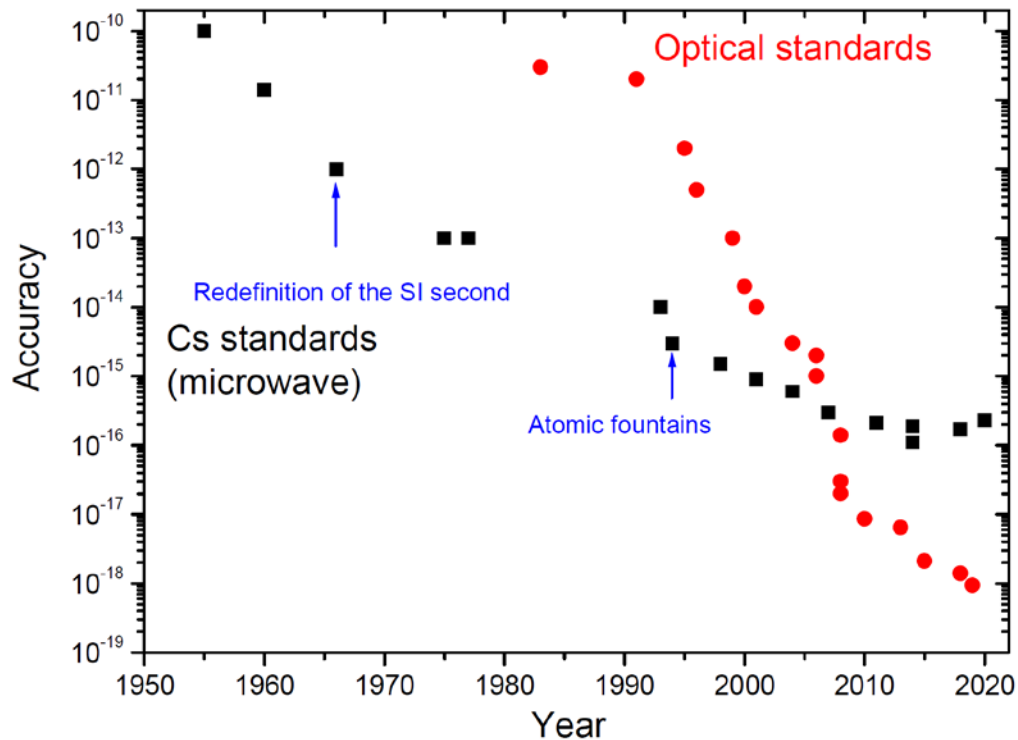


Figure 1 - Evolution of the relative frequency uncertainty of frequency standards

In order to prepare for a future redefinition of the SI second, the Consultative Committee for Time and Frequency (CCTF) has investigated possible options and their impact on concerned communities and applications, not just scientific and technological, but also regulatory and legislative. This was carried out through an online survey from December 2020 to January 2021. In addition, a dedicated CCTF Task Force has been created and subdivided into three subgroups related to:

- Requests from user communities, National Metrology Institutes (NMIs) and liaisons
- Atomic frequency standards, and possible redefinition approaches
- Time and frequency dissemination and time scales

The survey found that in terms of awareness and interest of the user communities in the possible new definition, overall, the time and frequency (T/F) community is well informed about the redefinition efforts and is supportive of the initiative. Several of the survey's responders from NMIs, academia and to industry indicated that they would benefit from access to optical frequency standards and more accurate T/F signals associated with the redefinition. The survey responses also revealed that many industrial users have current and future T/F needs that can be addressed by the current definition and dissemination methods. The redefinition campaign presents an opportunity for CCTF members and timing laboratories to improve and enhance their efforts in education and knowledge transfer of the T/F metrology principles and techniques. The survey responses indicated that limited effort, if any, would be required to adapt existing laws and regulations to the new definition. This aspect would require additional consultation and validation to ensure that formal administrative and legal processes are launched in a timely manner so as not to hinder the redefinition schedule.

In order to choose the best new definition and its implementation timeline, and to provide the CGPM with all the required information to make its decision, criteria and conditions have been defined to ensure that the following goals of the redefinition are reached:

- Offer an improvement by 10 to 100 of the realization of the new definition in the short term after the redefinition (reaching 10^{-17} to 10^{-18} relative frequency uncertainty) and potentially a larger improvement in the longer term;
- Ensure continuity with the current definition based on caesium;
- Ensure continuity and sustainability of the availability of the new SI second through TAI/UTC, and enable a significant improvement of the quality of TAI as soon as the definition is changed;
- Enable the dissemination of the unit to a broad category of users;
- Be acceptable by all NMIs and stakeholders.

In 2016, the CCTF Working Group on Strategic Planning (WG-SP) prepared a first version of the roadmap and the associated conditions for the redefinition [1]. Since June 2020, the overall roadmap has been updated by a dedicated CCTF Task Force on this topic, including the update of criteria and conditions that quantify the status of the developments and their maturity for a redefinition. The conditions listed in the 2016 document have been maintained, with some clarification where needed, and integrated into an updated list of mandatory criteria, that needs to be fulfilled before redefinition. For instance, new criteria relative to relativistic geodesy requirements and the acceptability of the new definition have been introduced.

Ancillary conditions for the redefinition have also been added: they need to achieve an advanced status, even if not completely fulfilled at the time of the redefinition.

Following the presentation of the new version of criteria and conditions at CCTF session 22-1 (October 2020), the Task Force has analysed the feedback from CCTF members, observers, liaisons and working groups. Corrections and improvements have been incorporated and were presented at CCTF session 22-2 (March 2021).

2. Possible options and scenarios for the redefinition

a) Redefinition options

Three options for the redefinition of the SI second can be envisaged.

Option 1: New definition based on a single reference optical frequency

In this option, the new definition is based on the approach of the present definition by choosing a radiation corresponding to a single atomic reference transition in the optical frequency range.

The primary realization is carried out with primary frequency standards based on this transition. Secondary representations of the second are provided by frequency standards based on other species (<https://www.bipm.org/en/publications/mises-en-pratique/standard-frequencies>). Caesium becomes a secondary representation of the SI second. The CCL-CCTF list of recommended frequency values continues to provide the means to investigate and validate promising new transitions.

Option 2: New definition based on an ensemble of reference optical frequencies

The second option proposes the use of the weighted geometric mean of an ensemble of chosen transition frequencies. The weight of each transition is initially fixed and inversely proportional to the squared uncertainty of the best standards based on this transition at the time of the redefinition [2].

The realization of the SI second is carried out with frequency standards based on transitions of part of the defined ensemble using a frequency ratio matrix updated by the CIPM [2]. Thus, each transition of the defined ensemble is a representation of the definition, including the current Cs reference transition if it is part of the ensemble. Such an approach optimally merges the concept and use of primary and secondary representations of the second. The CCL-CCTF list of recommended values continues to provide the means to investigate and validate promising new transitions.

The list of chosen transitions and their weights are periodically updated, including transitions that are already part of the ensemble or not yet part of it, and taking into account the evolution of the uncertainty of the best standards based on the chosen transitions.

Updates are carried out infrequently and only when the CIPM has validated any major progress of frequency standards and documented that the proposed update will bring a major improvement.

Option 3: New definition based on fixing the value of another fundamental constant

The third option aims at fixing the value of another fundamental constant, as it has been done for other SI units by fixing the value of the light speed c , Planck constant h , Boltzmann constant k , electron charge e . This approach is directly connected to the underlying fundamental framework: general relativity and the standard model of particle physics [3].

The *Mise en Pratique* would be based on atomic transition(s), either one as in Option 1 or an ensemble as in Option 2.

An important issue in the debate on a redefinition is to choose the best option for the new definition. At present, Option 3 is not achievable as there is no known fundamental physical constant with the necessary accuracy. However, there is still not a clear consensus on:

- i) the choice between Option 1 (single transition) and Option 2 (ensemble of transitions), even though the majority of participants in the survey favoured Option 1;
- ii) the identification of the preferable atomic radiation(s) for the definition.

Progress toward consensus will come from the following actions:

- The CCTF will clarify and make explicit the practical aspects of the new definition under every proposed option.
- The CCTF will increase the level of awareness of these practical aspects among CCTF members, the Consultative Committee for Units (CCU), International Committee for Weights and Measures (CIPM), General Conference on Weights and Measures (CGPM) and within a broader audience.
- Further development of optical frequency standards by NMIs and other institutes; achievements in terms of their operability; as well as further evaluation of specific redefinition alternatives, may provide a clearer choice of the redefinition option.

b) Redefinition scenarios

The possible redefinition scenarios depend on capabilities of optical frequency standards and their envisaged evolution, considering their performance, their readiness for sustainable contributions to the realization of time scales, especially UTC, and also their potential for commercial availability, and space qualification. A roadmap needs to address T/F transfer techniques required for the comparison of atomic clocks, for the construction of international time scales, and for the dissemination of reference signals to users, with an adequate uncertainty level.

Depending on the achievements and the development progress, the CCTF envisages three possible schedule options for the redefinition (Figure 2).

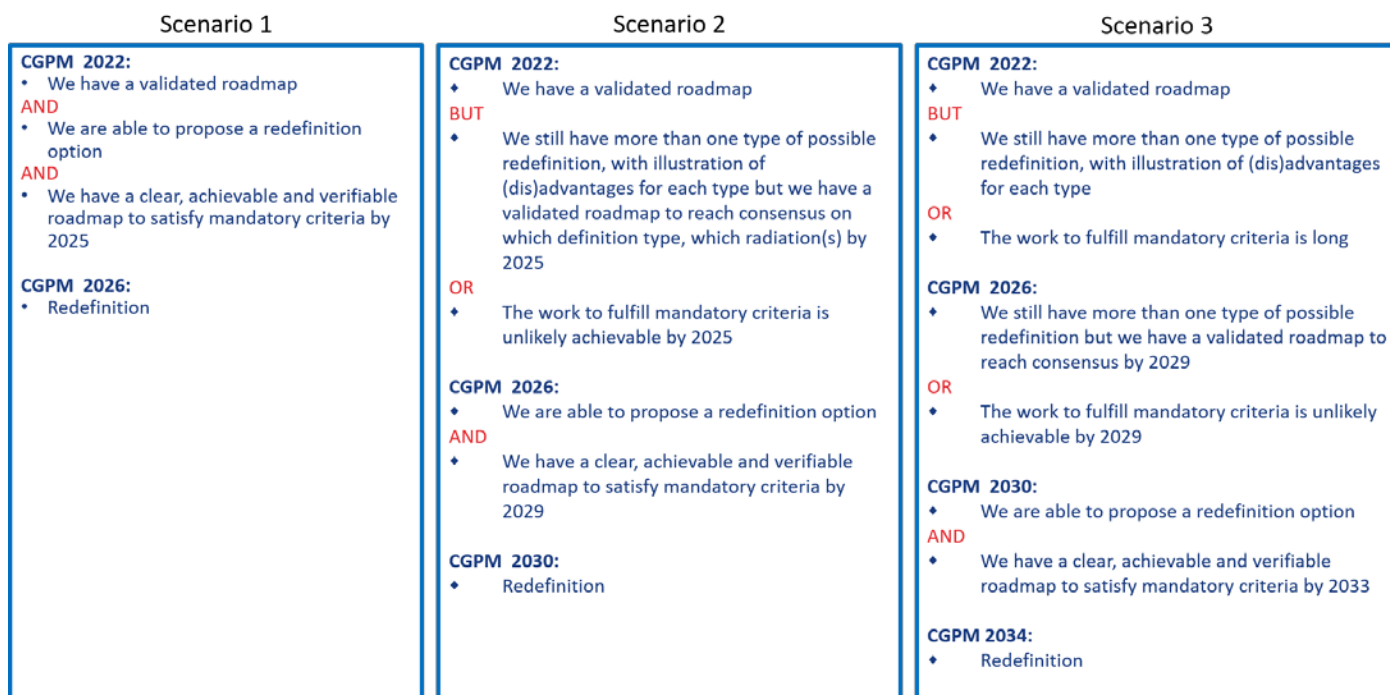


Figure 2 – Scenarios for the roadmap

The CCTF considers the first scenario leading to a redefinition in 2026 to be unrealistic. The maturity of the optical frequency standards, T/F transfer, and time scale formation is growing impressively. A considerable amount of work remains to be done before the redefinition in order to:

- choose the best option for the redefinition,
- develop, operate and compare optical frequency standards to provide regular contributions to TAI with secondary realizations of the second.

3. Priority setting of criteria and conditions to change the definition

The criteria and conditions that quantify the status of the development and the maturity for a redefinition according to options 1 and 2 are listed in the Table below.

Criteria and conditions related to:	Mandatory Criteria – Ancillary conditions
Frequency standards, including the contribution of Optical Frequency Standards to time scales	I.1 - Accuracy budgets of optical frequency standards I.2 - Validation of optical frequency standard accuracy budgets – frequency ratios I.3 - Continuity with the definition based on Cs I.4 - Regular contributions of optical frequency standards to TAI (as secondary representations of the second) I.5 - High reliability of OFS I.6 - Regular contributions of optical frequency standards to UTC(k)
T/F links for comparison or dissemination	II.1 – Availability of sustainable techniques for optical frequency standards comparisons II.2 – Knowledge of the local geopotential with an adequate uncertainty level II.3 – High reliability of ultra-high stability T/F links

Acceptability of the new definition	III.1 - Definition allowing future more accurate realizations III.2 – Access to the realization of the new definition III-3 - Continuous improvement of the realization and of time scales after redefinition III.4 - Availability of commercial optical frequency standards III.5 - Improved quality of the dissemination towards users
-------------------------------------	---

The criteria and conditions can be prioritized in two categories:

- the **mandatory criteria** that must be achieved before changing the definition (see Section 4) To be achieved for redefinition
- the **ancillary conditions** that need to be in an advanced status, even if not yet completely achieved at the time of the redefinition (see Section 5) Still in progress when redefinition

The boundary between these two categories, the so-called “mandatory achievements frontier”, defines the threshold for the criteria that must be necessarily fulfilled before taking the decision to change the definition.

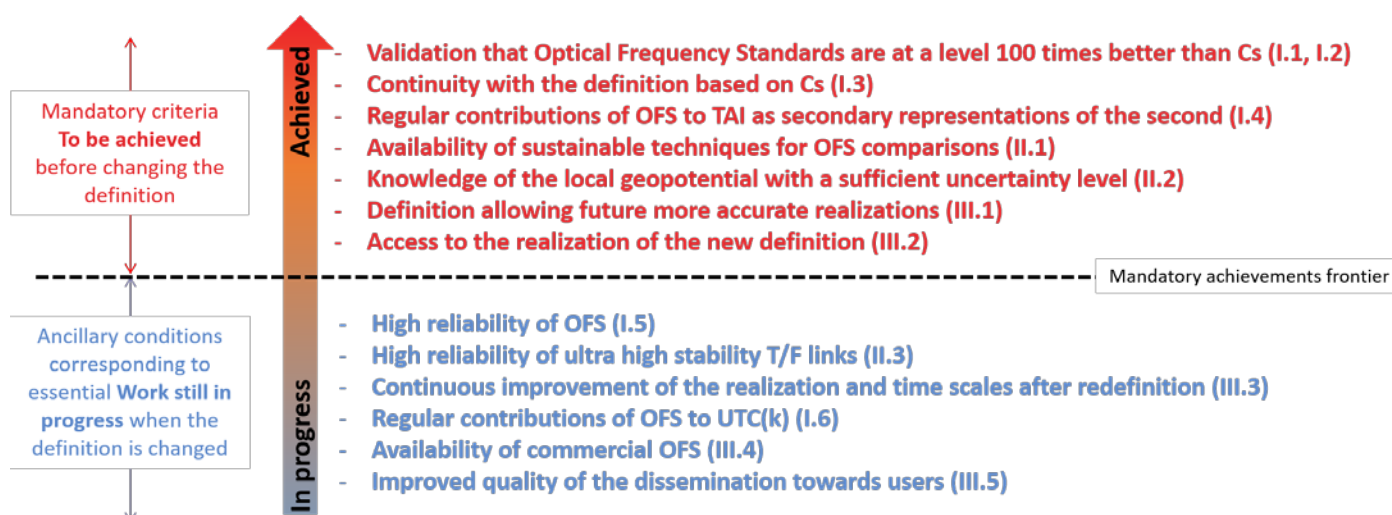


Figure 3 – Mandatory criteria and ancillary conditions to change the definition

The criteria, the conditions and their corresponding fulfilment indexes are detailed in Annex A for the three fields:

- frequency standards and their contribution to atomic time scales,
- techniques for T/F comparison or dissemination,
- acceptability of the new definition by NMIs and end-users.

4. Mandatory criteria

The mandatory criteria that must be achieved before changing the definition.

Validation that the relative frequency uncertainty of Optical Frequency Standards is at least 10-100 times better than Cs

The new definition must ensure an improvement by 10 to 100 in relative frequency uncertainty through its realization with Optical Frequency Standards, i.e. reaching 10^{-17} to 10^{-18} relative frequency uncertainty.

Corresponding criteria:

- *Criterion I.1 defining the minimum ensemble of identical and different OFS that demonstrate uncertainty budgets in the low 10^{-18} level*
- *Criterion I.2 ensuring the validation of these uncertainty budgets by direct comparisons or frequency ratio measurements*

Continuity with the definition based on caesium

The continuity between the current and the new definitions will be ensured by frequency evaluations, before the redefinition, of OFS that will contribute to the future realization of the new definition with respect to current realizations of the definition based on caesium (Cs fountains, TAI).

Corresponding criterion:

- *Criterion I.3 defining the minimum number and quality of OFS frequency evaluations with respect to TAI or Cs primary frequency standards*

Regular contributions of OFS to TAI (as secondary representations of the second)

After the redefinition, it will be important to ensure the continuity and sustainability of the availability of the redefined SI second through TAI/UTC, enabling a significant improvement in the quality of TAI as soon as the definition is changed.

Corresponding criterion:

- *Criterion I.4 defining the minimum number of regular OFS contributions to TAI before the redefinition (OFS acting as secondary representations of the second).*

Availability of sustainable techniques for Optical Frequency Standards comparisons

The capability to compare OFS with adequate uncertainty will be a key driver to validate OFS uncertainty budgets.

Corresponding criterion:

- *Criterion II.1 defining the level of availability and sustainability of transportable clocks or T/F links with appropriate uncertainties.*

Knowledge of the local geopotential with an adequate uncertainty level

In order to correct for the gravitational frequency shift affecting comparisons between distant frequency standards, the local geopotential must be known with an adequate uncertainty level (an uncertainty of 1 cm in the knowledge of the altitude of the OFS corresponds to a relative frequency uncertainty of 10^{-18}).

Corresponding criterion:

- *Criterion II.2 ensuring the appropriate knowledge of the local geopotential for the validation of OFS accuracy budgets, frequency ratio measurements and the calibration of TAI.*

Definition allowing more accurate realizations in the future

The new definition must be long lasting. In the short term (just after the redefinition), it must ensure an improvement by 10 to 100 of its realization with OFS, i.e. reaching 10^{-17} to 10^{-18} relative frequency uncertainty. In the longer term, it must also allow further improvements of the realizations in order to avoid any early obsolescence of the definition.

Corresponding criterion:

- *Criterion III.1 ensuring that the new definition has the potential to lead to long-term improvement of the realization beyond 10^{-18}*

Access to the realization of the new definition

The new definition must be acceptable by all NMIs, either those that will have the capability to develop their own primary or secondary OFS, or not. Thus, NMIs and high accuracy users must have access to primary or secondary realizations of the new definition, and the *Mise en Pratique* of the new definition must be easily understandable with a clear uncertainty evaluation process. Moreover, to avoid any significant impact on NMIs using only Cs as reference clocks, Cs frequency standards will continue to realize the new definition as secondary frequency standards. UTC will continue to provide a realization of the SI second to all NMIs and users.

Corresponding condition:

- *Criterion III.2 ensuring that the realization / “Mise en Pratique” of the new definition will be easily understandable, with clear uncertainty evaluation process, that NMIs and high accuracy users will have access to primary or secondary realizations of the new definition, and that Cs frequency standards will ensure a secondary realization of the new definition.*

5. Ancillary conditions

Contrary to the mandatory criteria, ancillary conditions are not required to be fully achieved to change the definition. Nevertheless, they are important to ensure the best realization and exploitation of the new definition in the short and long term. Thus, these conditions correspond to essential work that must have started before the redefinition, with a reasonable amount of progress at the time of redefinition and a commitment of stakeholders to continue their efforts on the associated activities.

High reliability of optical frequency standards and ultra-high stability T/F links

The reliability and continuous operational capability of OFS and ultra-high stability T/F links will be key drivers to provide trusted realizations of the SI second that fully exploit the OFS potential and to contribute efficiently to the calibration of TAI.

Corresponding conditions:

- *Condition I.5 ensuring a reliable continuous operational capability of OFS over durations > 10 days, in the laboratory environment.*
- *Condition II.3 ensuring on-demand continuous operational capability of T/F links over sufficient durations that do not limit OFS comparisons and their regular contributions to TAI.*

Continuous improvement of the realization and of time scales after redefinition

The new definition will likely offer strong potential for improvement in the short and long term, both for its realization and for the quality of time scales. Long-term improvements will require efforts from NMIs to push OFS accuracy limits further, to develop new OFS and to maintain Cs primary frequency standards as long as their contributions will be necessary.

Corresponding conditions:

- Condition III.3 ensuring the commitment of NMIs to undertake their best efforts to:
 - o improve and operate optical frequency standards that provide primary or secondary realizations of the new definition (reliable / continuous operation, regular contributions to TAI, ...).
 - o maintain the operation of Cs fountain standards over the appropriate duration.

Widest adoption of the new definition

In order to ensure the widest acceptance and to take best advantage of the new definition by stakeholders and NMIs that will not develop their own OFS, efforts must be made to develop commercial optical frequency standards, and to improve the quality of the dissemination towards users. This has already been illustrated over decades with commercial Cs clocks and T/F links (GNSS, TWSTFT, fibre / internet). It is expected that the progress in these domains will be boosted with the change of definition.

Corresponding conditions:

- Condition III.4 ensuring the availability of commercial optical frequency standards.
- Condition III.5 ensuring an improved quality of the dissemination towards users.

Regular contributions of optical frequency standards to UTC(k)

The use of Cs fountains in the steering of some UTC(k)s has led to a dramatic improvement of both accuracy and long-term stability of the concerned UTC(k). It is likely that the integration of OFS in this steering will improve further the quality of the UTC(k) produced by the concerned NMIs.

Corresponding condition:

- Condition I.6 ensuring the regular contribution of optical frequency standards to UTC(k).

6. Fulfilment levels of criteria and conditions

The fulfilment of the criteria and conditions are continuously monitored by the CCTF through its WGs, and periodically assessed by the CCTF WG on Strategic Planning by the use of “fulfilment indexes” that help to:

- evaluate the current status of fulfilment and progress,
- be aware of the remaining work to fulfil all mandatory criteria,
- be able to quantitatively follow the improvements,
- decide it is time to change the definition.

Different indicator types have been defined (see Annex A):

- quantitative fulfilment indexes for the mandatory criteria,
- qualitative progress indexes for the ancillary conditions.

a) Fulfilment levels of mandatory criteria

The fulfilment levels of mandatory criteria are shown in Figure 4. Fulfilment regions have been defined, from very low fulfilment levels (0 %-40 %, region in red) to satisfactorily fulfilment levels (80 %-100 % and above, region in green). The dashed grey line in Figure 4 defines the threshold above which the criteria can be considered as fulfilled.

Mandatory criteria

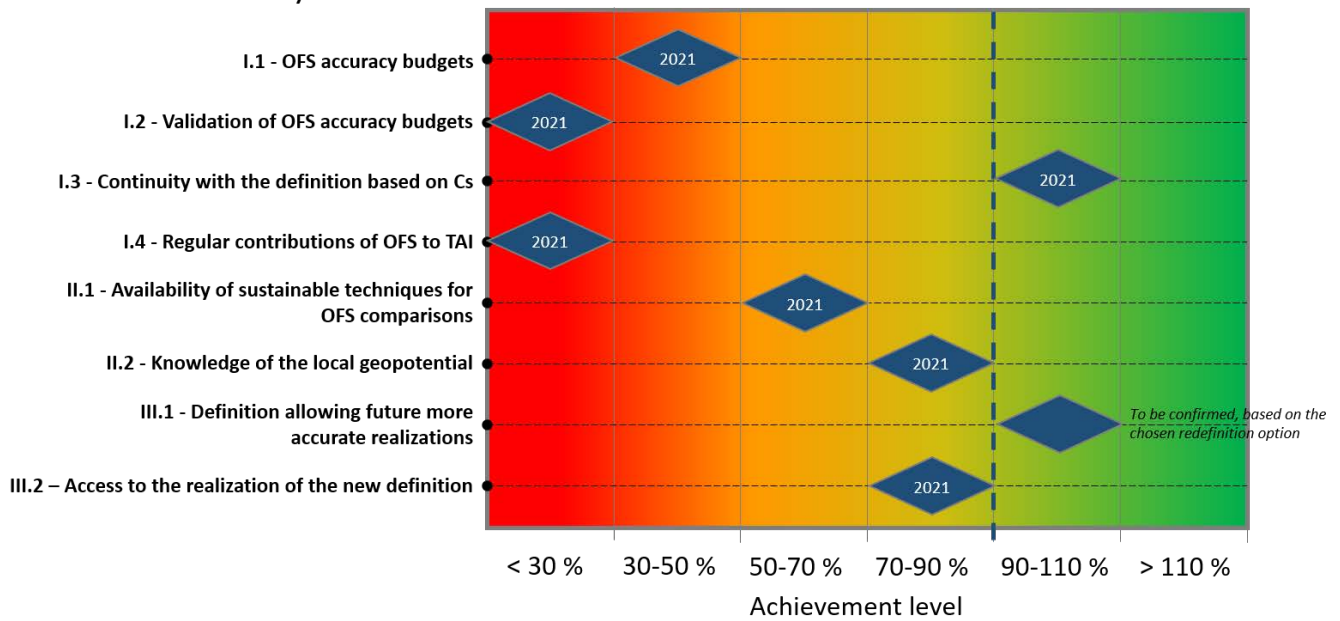


Figure 4 – Fulfilment levels of mandatory conditions

Situation in 2021:

While for certain criteria the fulfilment seems to have been almost achieved, for others the fulfilment is more challenging.

Due to the considerable number of optical frequency standards under development, good progress has been made on OFS performance Criteria I.1 and I.3 (fulfilment indices close to 50 % and 100 %, respectively) as well as on their realization viability Criteria III.1 and III.2 (fulfilment near 100 % for both, to be confirmed since they are based on the chosen redefinition option). However, the challenges associated with limited resources for developing multiple standards in one institute (along with limitations in long-distance time transfer) have led to a low fulfilment index for the optical frequency standards' comparison Criterion I.2 (fulfilment < 30 %). In addition, demands associated with long measurement runs at high duty cycle and prioritizing the work on other criteria have slowed progress on the OFS contributions to TAI Criterion I.4 (fulfilment < 30 %).

The fulfilment of the criterion II.2 related to the knowledge of the geopotential is achieved in the majority of NMIs that operate an OFS.

For the criteria II.1, a sustainable technique for OFS comparison at the proper uncertainty level is more challenging. Over an intracontinental scale (baselines of about 1000 km), the requirement is fulfilled by optical fibre links, even if a significant effort for regular comparison campaigns should be addressed.

b) Progress levels of ancillary conditions

The progress status of developments with respect to these ancillary conditions is shown in Figure 5.

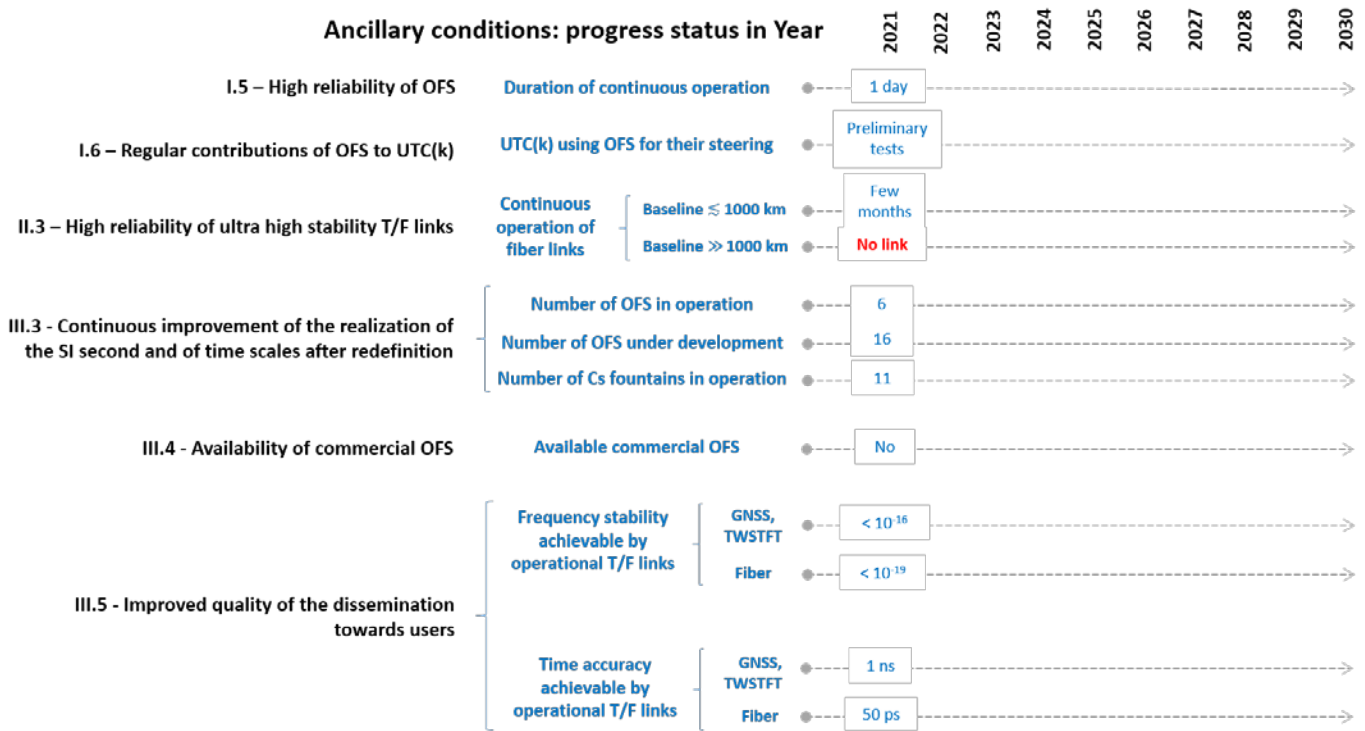


Figure 5 – Progress status of developments with respect to ancillary conditions

Situation in 2021:

Condition II.3, OFS comparisons are presently possible via reliable optical fibre links over continental scales (baselines of about 1000 km). However, a significant effort for regular comparison campaigns should still be addressed. Conversely, comparisons over intercontinental scales (baselines much longer than 1000 km) at the required uncertainty are not presently available. Optical fibre links and transportable OFS are viable means, but to date no intercontinental OFS comparisons with fibre links or with transportable OFS have been demonstrated. Only one single intercontinental fibre link has been demonstrated so far (without an OFS comparison). Hence, significant efforts should be addressed i) to ensure regular comparisons of OFS at the continental scale, where current possibilities fulfil the requirements; ii) to provide realistic and reliable means to compare OFS at the intercontinental scale. Research is also progressing on novel T/F transfer techniques (VLBI, optical TW communication link on satellites, ...). The necessary level of accuracy and reliability has not yet been achieved, but the anticipated evolution in the next few years is promising and additionally is under investigation also for commercial purposes in other communities. The complete achievement will probably be possible only over the long term, possibly a few decades.

Condition I.6, preliminary experiments have been carried out to evaluate the impact of integrating an OFS in the steering of a UTC(k). However, no UTC(k) is presently steered by an OFS in a fully operational mode.

Condition III.4, no atomic frequency standards operating in the optical domain are commercially available in 2021.

Condition III.5, the high precision operational dissemination towards users can be carried out either by satellite techniques (GNSS, TWSTFT, ...) or ground-based fibre links. Today, only fibre links have the capability to transfer OFS frequencies without degradation.

7. Conclusion

The goal to achieve a better definition and realization of the SI second is intended to serve current and future needs in metrology and to foster scientific and technological applications at the highest accuracy.

The CCTF, having examined the necessary criteria to be achieved before changing a definition, having consulted users and stakeholders, and particularly having thoroughly studied the necessary achievements in the development of frequency standards, their comparison techniques, and their capacity to contribute to time scales, considers the schedule scenario aiming at a redefinition in 2026 to be unrealistic.

The maturity of the optical frequency standards, T/F transfer, and time scale formation is growing impressively and there is a clear way forward to choose between the two options for the new definition: a single atomic reference transition in the optical frequency range or an ensemble of reference transitions.

Therefore, the CCTF recommends adopting a roadmap towards the redefinition of the SI second according to the criteria and ancillary conditions listed in this document, and preferably according to the schedule scenario aiming at a redefinition in 2030, with a back-up schedule scenario leading to a redefinition in 2034.

ANNEX A - Details on mandatory criteria and ancillary conditions - Fulfilment and progress indexes

I - Criteria and conditions related to frequency standards and their contribution to time scales

II - Criteria and conditions related to T/F links for comparison or dissemination

III - Criteria and conditions related to the acceptability of the new definition

I - Criteria and conditions related to frequency standards and their contribution to time scales

To be achieved
for redefinition

Criterion I.1 - Accuracy budgets of optical frequency standards

At least three optical frequency standards based on the same reference transition, in different institutes, have demonstrated evaluated relative frequency uncertainties $\lesssim 2 \times 10^{-18}$ (*).

At least three frequency evaluations of optical frequency standards based on different reference transitions, either in the same institute or different institutes, have demonstrated evaluated uncertainties $\lesssim 2 \times 10^{-18}$ (*).

(*) *Based on comprehensive, comparable and published accuracy budgets.*

Fulfilment Indexes:

- Number of OFS based on the same reference transition, in different institutes, with evaluated uncertainties $\lesssim 2 \times 10^{-18}$

Fulfilment level:

- **Up to June 2021: 20-40 %**

Sr: 2.0×10^{-18} [4], 5.5×10^{-18} [5]

Yb: 1.4×10^{-18} [6]

Yb+: 3.2×10^{-18} [7], 2.7×10^{-18} [9]

Al⁺: 9.0×10^{-19} [8]

Ca⁺: 3.0×10^{-18} [10]

- Number of frequency evaluations of OFS based on different reference transitions, either in the same institute or different institutes, with evaluated uncertainties $\lesssim 2 \times 10^{-18}$

Fulfilment level:

- **June 2021: 80-100 %**

Three different reference transitions (in different laboratories) have been evaluated at the required level:

Sr (2.0×10^{-18}) [4], Yb (1.4×10^{-18}) [6], Al⁺ (9.0×10^{-19}) [8].

→ **June 2021: Overall Fulfilment Index for criterion I.1: 30-50 %**

Index monitored by CCTF-WGPSFS, CCL-CCTF-WGFS

Criterion I.2 - Validation of Optical Frequency Standard accuracy budgets – Frequency ratios

- Unit ratios (frequency comparison between standards with the same clock transition): at least three measurements between OFS in different institutes with an agreement $\Delta\nu/\nu \lesssim 5 \times 10^{-18}$ (either by transportable clocks or advanced links). Applicable to at least one radiation of I.1.
- Non-unit ratios (frequency comparison between standards with different clock transitions): at least five measurements between standards among I.1 or other, each ratio measured at least twice by different institutes with an agreement $\Delta\nu/\nu < 5 \times 10^{-18}$ (either by direct comparisons, transportable clocks or advanced links).

Fulfilment Indexes:

- Number of measurements of OFS with same clock transition in different institutes with an agreement $\Delta\nu/\nu \lesssim 5 \times 10^{-18}$

Fulfilment level:**- June 2021: 20-40 %**

Unit ratios for three different reference transitions from I.1) have been measured once within the stated uncertainty: Yb (9×10^{-19}) [6], Yb⁺ (4.2×10^{-18}) [9], Sr (4.7×10^{-18}) [5]

- Number of measurements of optical frequency ratios between standards of I.1 or other, each ratio measured at least twice by independent institutes with an agreement $\Delta\nu/\nu < 5 \times 10^{-18}$

Fulfilment level:**- June 2021: 0-20 %**

Al⁺/Yb 4.8×10^{-18} [11]

Al⁺/Sr 7.0×10^{-18} [11]

Yb/Sr 6.4×10^{-18} [11]

→ June 2021: Overall Fulfilment Index for Criterion I.2: < 30 %

Indexes monitored by CCTF-WGPSFS, CCL-CCTF-WGFS

Criterion I.3 - Continuity with the definition based on Cs

There are at least three independent frequency evaluations of the optical frequency transitions utilized by the standards in I.1) with TAI or with three independent Cs primary frequency standards (in different or the same institutes), possibly also via optical frequency ratio measurements, where the measurements are limited essentially by TAI or by the uncertainty of these Cs frequency standards ($\Delta\nu/\nu < 3 \times 10^{-16}$).

Fulfilment Index:

- Number of independent frequency evaluations of OFS listed in I.1) with uncertainty $\Delta\nu/\nu < 3 \times 10^{-16}$

Fulfilment level:**- June 2021: 90-110 %**

Evaluations at the stated level:

Sr: Five evaluations (published between 2013 and 2020) with an uncertainty in the range $1.5 - 3.1 \times 10^{-16}$ [12], [13], [14], [15], [16]

Yb: Two evaluations (published in 2019 and 2020) with an uncertainty in the range $2.1 - 2.6 \times 10^{-16}$ [17], [18]

Yb+(E3): One frequency evaluation (published in 2021) with an uncertainty of 1.3×10^{-16} [19]

Indexes monitored by CCTF-WGPSFS, CCL-CCTF-WGFS

Criterion I.4 - Regular contributions of optical frequency standards to TAI (as secondary representations of the second)

At least three state-of-art calibrations of TAI (uncertainty $\lesssim 2 \times 10^{-16}$ without counting the recommended uncertainty of the secondary representation of the second u_{srep}) each month from a set of at least five Optical Frequency Standards for at least 1 year. Check that there is no degradation of TAI if its calibrations were done by OFS considered as primary standards and Cs frequency standards considered as secondary standards.

Fulfilment Index:

- Number of calibrations of TAI each month from a set of at least five Optical Frequency Standards for at least 1 year

Fulfilment level:

- **June 2021:** < 30 % (less than 0.5 report per month); six OFS (three ^{87}Sr , three ^{171}Yb) have been validated by the WGPSFS for use in TAI steering (two in 2017, one in 2018, two in 2019, one in 2020). Ten new reports in total have been submitted, seven of which at a time concerned the current month and three at a time the previous month. Excluding the initial submission which concerns measurements made months earlier, “regular” participation in 2020 is at most 0.5 per month, and all with uncertainty $> 2 \times 10^{-16}$. None of the reports achieved the stated uncertainty, essentially because of Dead Time Uncertainty or because of Frequency Transfer Uncertainty or both.

Index monitored by CCTF-WGTAI

Condition I.5 – High reliability of OFS

Reliable continuous operation capability of OFS over durations > 10 days, in a laboratory environment, with the appropriate level of uncertainty.

Progress Index:

- Typical duration of OFS continuous operation

Progress status:

- June 2021: 1 day

Index monitored by CCTF-WGPSFS

Condition I.6 - Regular contributions of optical frequency standards to UTC(k)Progress Index:

- UTC(k) steered by OFS

Progress status:

- June 2021: Preliminary tests of UTC(k) steered by an OFS [20], [21], [22]

Index monitored by CCTF-WGTAI, CCTF-WGALGO

II - Criteria and conditions related to T/F links for comparison or dissemination

To be achieved
for redefinition

Criterion II.1 – Availability of sustainable techniques for Optical Frequency Standards comparisons

Availability and sustainability of transportable clocks or T/F links with uncertainties $< 5 \times 10^{-18}$ for frequency comparisons between at least NMIs operating optical frequency standards of I.1), on a national / intracontinental basis (baseline up to about 1000 km). Capability of repeated uncertainty estimations of these links.

Fulfilment Index:

- Availability of transportable clocks approaching 5×10^{-18} or T/F links with uncertainties $< 5 \times 10^{-18}$ on a national / intracontinental basis

Fulfilment level:

- June 2021: 50-70 %

Uncertainties of fibre T/F links $< 5 \times 10^{-18}$ (in Europe, Japan, ...)

Existence of transportable clocks approaching 5×10^{-18} [5]

Index monitored by CCTF-WGATFT, with contributions of CCTF-GNSS, CCTF-WGTWSTFT

To be achieved
for redefinition

Criterion II.2 – Knowledge of the local geopotential with an adequate uncertainty level

Knowledge of geopotential differences for NMIs operating OFS of I.2) to be consistent with the uncertainty budget of a frequency comparison between OFS using advanced links, i.e. including the uncertainty budget of the two OFS and of the link.

Knowledge of local geopotential for NMIs operating OFS of I.4) with an uncertainty corresponding to a frequency uncertainty $\lesssim 10^{-17}$, for the calibration of TAI.

Fulfilment Index:

- For comparisons between OFS of I.2), uncertainty in the geopotential difference values sufficiently low with respect to the required uncertainty

-For NMIs operating OFS of I.4) for the calibration of TAI, uncertainty in the geopotential value $\lesssim 10^{-17}$

Fulfilment level:

- June 2021: 70-90 %

(Uncertainty of geopotential: NIST: 6×10^{-18} , INRIM: 3×10^{-18} , SYRTE: 4×10^{-18} , PTB: 2×10^{-18} , NICT: 2×10^{-17} , KRISS: 3×10^{-17} , NPL: 3×10^{-18} , NMIIJ: 6×10^{-17})

[23], [24], [25], [26], [27], [28]

Index monitored by CCTF-WGPSFS, with contributions of CCTF-WGATFT, CCTF-WGTAI

Still in progress
when redefinition

Condition II.3 – High reliability of ultra-high stability T/F links

On-demand continuous operation capability of T/F links over sufficient durations that do not limit OFS comparisons and their regular contributions to TAI.

Progress Index:

- Existence of reliable ultra-high stability T/F links

Progress status:

- June 2021: A few months continuous unmanned operation of fibre links for intracontinental comparisons [29], [30]

No existing link that does not limit OFS intercontinental comparisons

Index monitored by CCTF-WGATFT, with contributions of CCTF-GNSS, CCTF-WGTWSTFT

III – Criteria and conditions related to the acceptability of the new definition

To be achieved
for redefinition

Criterion III.1 - Definition allowing more accurate realizations in the future

The new definition must be long lasting. In the short term (just after the redefinition), it must ensure an improvement by 10/100 of its realization with OFS, i.e. reaching 10^{-17} / 10^{-18} relative frequency uncertainty. In the longer term, it must have the potential for further improvement of the realization of 10^{-18} and beyond in order to avoid any early obsolescence of the definition.

Fulfilment Index:

- No identified fundamental effect limiting OFS accuracy at the 10^{-18} level

Fulfilment level:

- **June 2021: 100 %** (To be confirmed, based on the chosen option for the redefinition, but all species in I.1, and some newer systems have the potential to go beyond 10^{-18})

To be achieved
for redefinition

Criterion III.2 - Access to the realization of the new definition

- Realization / “*mise en pratique*” of the new definition must be easily understandable with a clear uncertainty evaluation process;
- Access for NMIs and high accuracy users to primary or secondary realizations of the new definition;
- Cs frequency standards ensure a secondary realization of the new definition.

Fulfilment Indexes:

- Existence of a document explaining the *Mise en Pratique* of the definition

Fulfilment level:

- **June 2021: 0 %** (No existing document; pending the choice of the redefinition option)

- Access for NMIs and high accuracy users to primary or secondary realizations of the definition at least similar to the current situation

Fulfilment level:

- **June 2021: 100 %** (To be confirmed, based on the chosen option for the redefinition, but primary or secondary representations of the SI second will continue to be accessible via metrology institutes or TAI)

- Insurance of the capability of using Cs frequency standards as secondary realization of the new definition

Fulfilment level:

- **June 2021: 100 %** (existing TAI architecture will be maintained at current level or better and Cs will be a secondary representation of the second.

→ **June 2021: Overall Fulfilment Index for Criterion III.2: 70-90 %**

Condition III.3 - Continuous improvement of the realization and of time scales after redefinition

Commitment by NMIs to make the best effort to:

- improve and operate optical frequency standards that provide primary or secondary realizations of the new definition (reliable / continuous operation, regular contributions to TAI, ...);
- develop new OFS;
- maintain the operation of Cs fountain standards over the appropriate duration.

Progress index:

- Number of operating OFS, number of OFS under development, number of operating Cs fountains

Progress status:

- June 2021: eleven Cs fountains in operation, 22 OFS (six in operation, 16 under development) with the following species: Sr (nine), Yb (five), Yb+ (four), Hg (one), Al+ (one), Sr+ (one), In+ (one)

Index monitored by CCTF-WGPSFS

Condition III.4 - Availability of commercial optical frequency standards

Progress index:

- Available commercial OFS

Progress status:

- June 2021: No available commercial OFS

Index monitored by CCTF-WGPSFS, CCTF-WGTAI

Condition III.5 - Improved quality of the dissemination towards users

Progress index:

- Performance of T/F links (GNSS, TWSTFT, fibre / internet) for the dissemination of the definition towards users

Progress status:

- Frequency stability:

- June 2021: 10^{-17} – 10^{-16} for satellite microwave techniques (GNSS, TWSTFT); 10^{-20} level for fibre links [31]

- Time accuracy:

- June 2021: 1 ns for satellite microwave techniques (GNSS, TWSTFT); 50 ps for fibre links [32]

Index monitored by CCTF-WGTAI, CCTF-GNSS, CCTF-WGTWSTFT, CCTF-WGATFT, CCTF-WGMRA

ANNEX B - References

- [1] CCTF strategy document, 2016, <https://www.bipm.org/documents/20126/35554894/CCTF+Strategy/7cf0f648-2afe-d15c-0909-1f03406bbb8f>, <https://www.bipm.org/documents/20126/35554967/CCTF-roadmap.jpg/6338cf4c-59d6-d8ee-6cc5-eb390715b900>
- [2] J. Lodewyck, *Metrologia* **56**, 055009 (2019)
- [3] C. Bordé, *C. R. Physique* **20**, 22 (2019)
- [4] T. Bothwell, D. Kedar, E. Oelker, J. M. Robinson, S. L. Bromley, et al., *Metrologia* **56**, 065004 (2019)
- [5] M. Takamoto et al., 2021, *Nature Photonics* **14** 411-415
- [6] W. F. McGrew, X. Zhang, R. J. Fasano, S. A. Schäffer, K. Beloy, et al., *Nature* **564**, 87 (2018)
- [7] N. Huntemann et al. *Phys. Rev. Lett* **116** 063001 (2016)
- [8] S. M. Brewer, J.-S. Chen, A. M. Hankin, E. R. Clements, C. W. Chou, et al., *Phys. Rev. Lett.* **123**, 033201 (2019).
- [9] Christian Sanner, Nils Huntemann, Richard Lange, Christian Tamm, Ekkehard Peik, Marianna S. Safronova & Sergey G. Porsev, *Nature* **567**, 204 (2019).
- [10] Yao Huang , Baolin Zhang, Mengyan Zeng, Yanmei Hao , Huaqing Zhang, Hua Guan, Zheng Chen, Miao Wang, & Kelin Gao, arXiv:2103.08913.
- [11] K. Beloy, M. I. Bodine, T. Bothwell, S. M. Brewer, S. L. Bromley, et al., *Nature* **591**, 564 (2021)
- [12] R. Le Targat et al., *Nat. Comm.* **4**, 2109 (2013)
- [13] J. Lodewyck et al., *Metrologia* **53**, 1123 (2016)
- [14] C. Grebing et al., *Optica* **3**, 563 (2016)
- [15] R. Schwarz et al., *Phys. Rev. Reas.* **2**, 033242 (2020)
- [16] N. Nemitz et al., *Metrologia* **58**, 025006 (2020)
- [17] W. McGrew et al., *Optica* **6**, 448 (2019)
- [18] M. Pizzocaro et al., *Metrologia* **57**, 035007 (2020)
- [19] R. Lange et al., *Phys. Rev. Lett.* **126**, 011102 (2021)
- [20] H. Hachisu, F. Nakagawa, Y. Hanado and T. Id, *Scientific Reports* **8**, 4243 (2018)
- [21] J. Yao et al. 2019, *Phys. Rev. Applied* **12**, 044069
- [22] W. R. Milner et al. 2019, *Phys. Rev. Lett.* **123**, 173201
- [23] <https://www.bipm.org/en/time-ftp/data>
- [24] C. Lisdat et al., *Nature Comm*, 12443, 7 , (2016)
- [25] Grotti, J. et al., *Nat Phys*, 437 , 14 (2018)
- [26] Denker H, et al., *J. Geod.* **92** 487–516 (2018)
- [27] Nikolaos K Pavlis and Marc A Weiss 2017 *Metrologia* **54** 535
- [28] Riedel et al, *Metrologia* **57**, 045005 (2020)
- [29] F. Guillou-Camargo, et al., *Applied Optics* **57**, 7203-7210 (2018)
- [30] Cantin, E. et al., *New J. Phys.* **23**, 053027 (2021)
- [31] Raupach, S. M. F., Koczwara, A. & Grosche, *Physical Review A* **92**, 021801 (2015)
- [32] Ł Śliwczyński et al 2019 *Metrologia* **56** 015006